

Deliverable Sentence: I used a breakpoint to catch the null state before `.map()` crashed the page, React DevTools to find the `inStock` prop typo that silently broke the badges with zero console errors, and the Network tab to spot the 404 URL typo that the console alone couldn't explain.

DEBUGGING JOURNAL (THE 3 BUGS)

Bug 1: App Crash (.map on null)

Symptom: Blank screen crash when rendering products.

Tool: Chrome Sources Breakpoint.

Showed: Paused on `.map()`; Scope pane showed products was null.

Fix: Initialized to `[]` and added `(products ?? []).map(...)`.

Bug 2: Silent Prop Typo

Symptom: Cards always said "Sold Out", console had 0 errors.

Tool: React DevTools (Components).

Showed: Prop passed as `isAvailable` instead of `inStock` (was undefined).

Fix: Passed `inStock={product.inStock}` to match props interface.

Bug 3: Network 404 Error

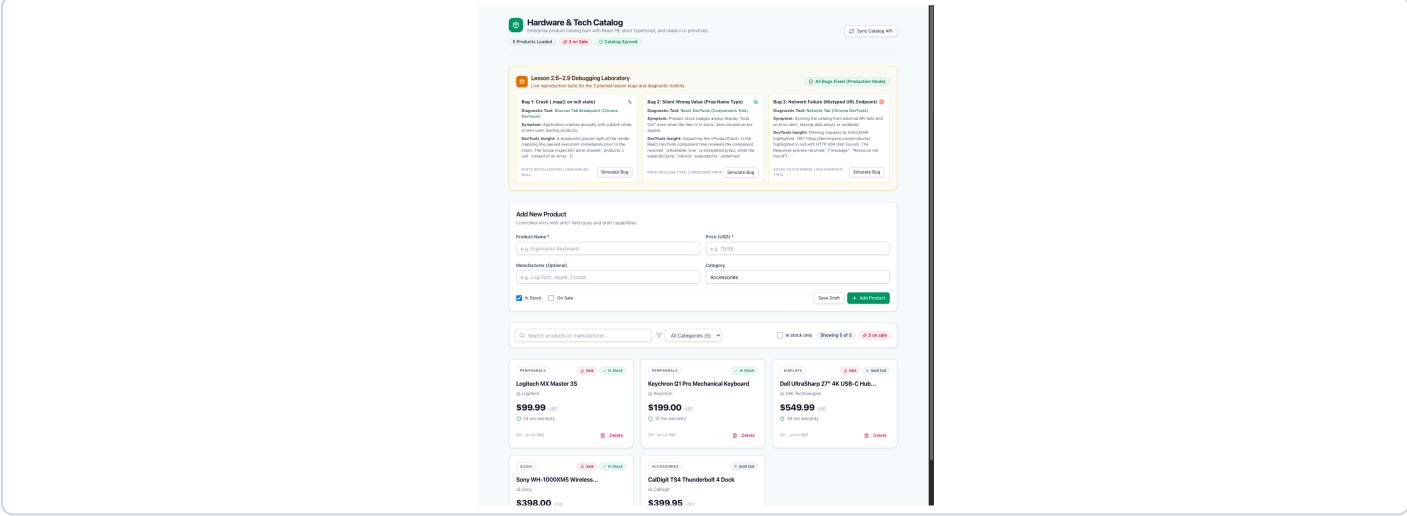
Symptom: Sync API failed with red error alert.

Tool: Chrome Network Tab (Fetch/XHR).

Showed: Red 404 on `/productss` due to extra 's'. Response: resource not found.

Fix: Corrected endpoint to `/products` and added `response.ok` check.

1. CATALOG OVERVIEW WITH SHADCN UI PRIMITIVES & TYPED BADGES



2. GIT COMMITS SPLIT PER REQUIREMENT

```
~/techpreneur/react/cheapanharith_Task-2-Typed-Debugged-Lessons-2.6-2.9 > git log --oneline
6629f35 (HEAD -> main, origin/main) docs: add screenshots and submission pdf
fb70475 fix: resolve planted bugs with DevTools tools and document in DEBUGGING_JOURNAL.md
4cf6957 test: plant 3 lesson bugs (crash on null map, prop typo silent failure, network 404)
ac5780d feat: derive types with Omit, Partial, and optional chaining with nullish coalescing
15306ca feat: type mini-app end-to-end with strict props interfaces and zero any
972455e feat: initialize Vite + React 19 + TypeScript project with Tailwind and @/components/ui
```

3. TYPESCRIPT CLEAN COMPILATION & PRODUCTION BUILD VERIFICATION

```
~/techpreneur/react/cheapanharith_Task-2-Typed-Debugged-Lessons-2.6-2.9 on (main)
> npx tsc --noEmit
✓ Done without errors. [0 diagnostics reported]

> npm run build
> cheapanharith-task-2-typed-debugged-lessons-2-6-2-9@0.0.0 build
> tsc -b && vite build
✓ 1892 modules transformed.
dist/index.html 0.53 kB | gzip: 0.35 kB
dist/assets/index-1VUnmkyK.css 23.32 kB | gzip: 5.36 kB
dist/assets/index-DKDFcxak.js 281.35 kB | gzip: 87.92 kB
✓ built in 100ms with 0 type errors
```